GATE

ALL BRANCHES



General Aptitude

QUANTITATIVE APTITUDE

Lecture No.- 03



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Recap of Previous Lecture



Topic Clocks

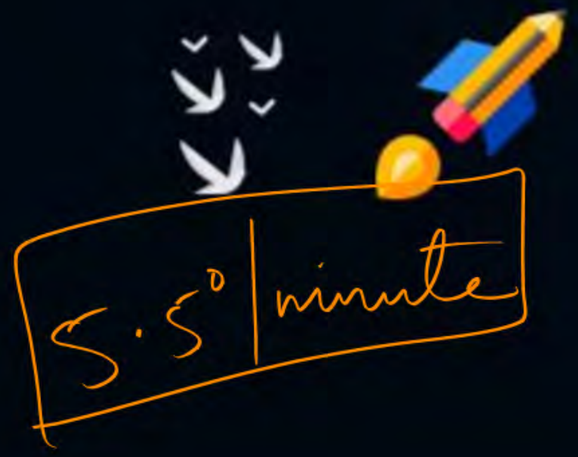
Time → Angle

Topics to be Covered



Topic-1 More on Clocks

Topic-2 Average



[MCQ]

#Q. At 9:45, the two hands of a clock make an angle of?

Assignment

A 60°

B 45°

C 33 ⅓°

D 22 ½°

$9 \Rightarrow 270°$

$45 \times 5.5 \Rightarrow 247.5$

$337.5°$

$22.5°$

Second Pattern:

Angle $\Rightarrow$ Time ?

0° OR Coincide:

12 hours ⇒ 11 times

24 hours ⇒ 22 times



180° OR Opposite:

Except 0° & 180°

1 hr ⇒ 2 times

12 hours = 11 times

24 hours = 22 times

12 times



90° OR Right angle:

Any angle except 0° & 180°

12 hours = 22 times

24 hours = 44 times



Q. 2-3 @ 90°

Questionnaire:

$2 : 27\frac{3}{11}$

#Q. In between 2 O' clock and 3 O' clock at what time the hands of clock form 90°?

$\frac{150}{5.5} \times 2 = \frac{300}{11}$

$= 27\frac{3}{11}$ min



Questionnaire:

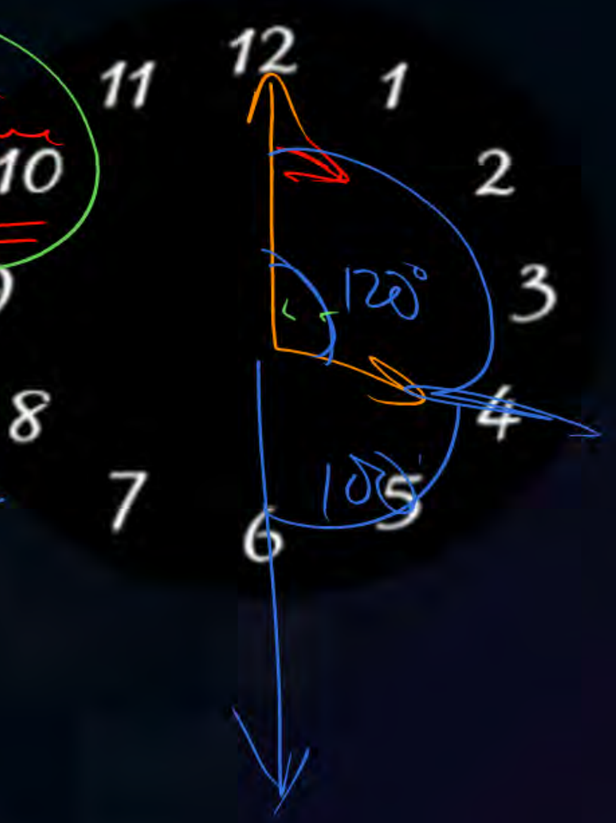
4:03 7/11

4:40

#Q. In between 4 0' clock and 5 0' clock at what time the hands of clock form 100°?

$\frac{20^{\circ}}{5.5} = \frac{40}{11} = 3\frac{7}{11}$ min

$\frac{220^{\circ}}{5.5} = \frac{440}{11} = 40$ min



Questionnaire:

$6:21\frac{9}{11}$

$6:43\frac{7}{11}$

#Q. In between 6 0' clock and 7 0' clock at what time the hands of clock form 60°?

$\frac{120}{5.5} = \frac{240}{11} = 21\frac{9}{11}$

$\frac{240}{5.5} = \frac{480}{11} = 43\frac{7}{11}$



Questionnaire:

$1:23\frac{7}{11}$

$1:52\frac{8}{11}$

#Q. In between 1 0' clock and 2 0' clock at what time the hands of clock form 100°?

260°

Reflex

$\frac{130}{5.5} = \frac{260}{11} = 23\frac{7}{11}$ min



$\frac{290}{5.5} = \frac{580}{11} = 52\frac{8}{11}$ min


260°

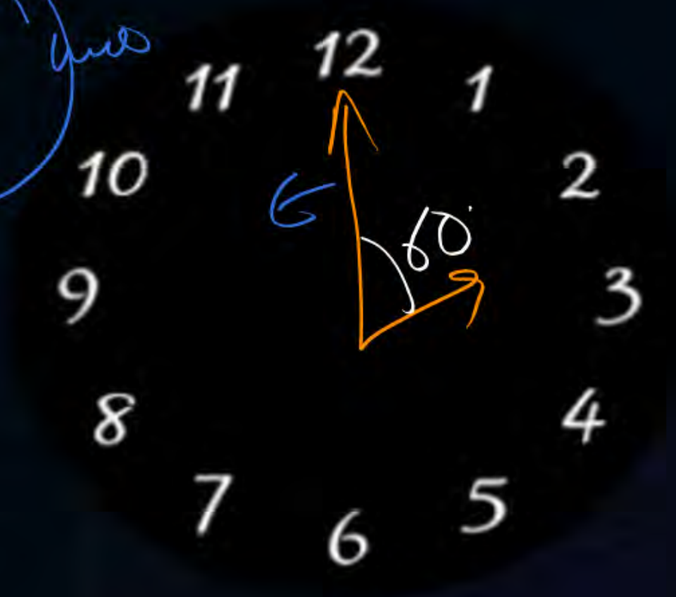

100°
260°

Questionnaire:

Q. 1 - 2 @ 100

$\frac{40}{5.5} = \frac{80}{11} = 7\frac{3}{11}$

$1:52\frac{8}{11}$



Third Pattern:

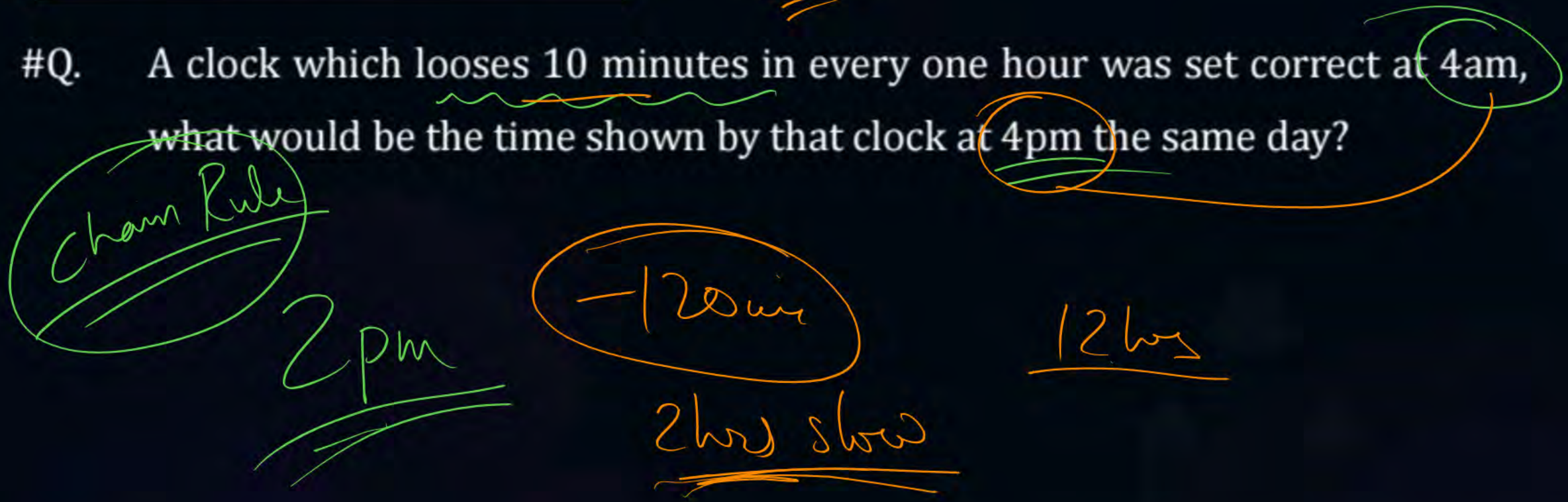
Gain OR logic

Questionnaire:

#Q. A Clock which gains 5 minutes in every one hour was set correct at 5am. What would be the time shown by that clock at 1pm the same day?

Questionnaire:

#Q. A clock which looses 10 minutes in every one hour was set correct at 4am, what would be the time shown by that clock at 4pm the same day?



MCQ
2 Marks

$6:21\frac{9}{11}$

PHYSICS WALLAH

#Q. At what time between 6AM and 7AM will the minute hand and hour hand of a clock make an angle closest to 60°?

$6:43\frac{7}{11}$

A 6:22 AM

B 6:27 AM

C 6:38 AM

D 6: 45 AM

$\frac{120}{5.5} = \frac{240}{11}$

$= 21\frac{9}{11}$



$\frac{240}{5.5} = \frac{480}{11}$

$= 43\frac{7}{11}$ min

#Q. A worker noticed that the hour hand on the factory clock had moved by 225 degrees during her stay at the factory. For how long did she stay in the factory?

A 3.75 hours

B 4 hours 15 minutes

C 8.5 hours

D 7.5 hours

450 min = $\frac{1^\circ}{2}$ min

$1^\circ \Rightarrow 2$ min

$225^\circ \Rightarrow 225 \times 2 = 450$ min

$\frac{450}{60}$ hr

$= 7.5$ hr

#Q. It is quarter past three in your watch. The angle between the hour hand and the minute hand is?

3:15

A 0°

B 15°

C 7.5°

D 22.5°

3 → 90°

15 × 3.5 → 82.5°

7.5°

AVERAGE



Normal Average ⇒ Mean

for Average ⇒ Mode

Middle Answer ⇒ Median

AVERAGE

$D = S \times T$

$S = \frac{D}{T}$

$T = \frac{D}{S}$

Average Speed



$$\frac{x}{20} + \frac{x}{30} = \frac{2x}{A.S}$$

$$\Rightarrow \frac{5x}{60_{12}} = \frac{2x}{A.S}$$

$$\therefore A.S. = 24 \text{ km/h}$$

AVERAGE

$$\text{Average} = \frac{\text{Sum of obs.}}{\text{No. of obs.}}$$

[MCQ]

$A = \frac{Sum}{No.}$

#Q. What would be the average of: 1, 2, 3, 4, 5, ...49, 50.

$S_n = \frac{n(n+1)}{2}$

Sum $\Rightarrow$ 1275 ?

$= \frac{n+1}{2}$

$x = 1 + 2 + 3 + 4 + \cdots\cdots 49 + 50$

$x = 50 + 49 + 48 + 47 + \cdots\cdots 2 + 1$

$2x = 50 \times 51$

$\frac{50+1}{2} = 25.5$

$x = \frac{50 \times 51}{2} = 1275$



2 mins Summary

Topic Average → A.M

$$\frac{n+1}{2}$$

THANK - YOU